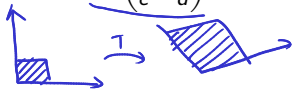


Diagonalization

$$\underbrace{\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix}}_{E^{-1}} \underbrace{\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}}_T \underbrace{\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}}_E = \underbrace{\begin{pmatrix} \exp(-i\theta) & 0 \\ 0 & \exp(i\theta) \end{pmatrix}}_D$$

determinant

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$



• A square matrix T is called diagonalizable if there exists an invertible matrix E such that $E^{-1}TE$ is a diagonal matrix D .

• The columns of E will be eigenvectors e_i of T .

• The corresponding diagonal entries in D will be eigenvalues λ_i of T .

• We solve the characteristic equation $\det(T - \lambda I) = 0$ for λ to get the eigenvalues of T .

• We use eigenvalue equation $Te_i = \lambda_i e_i$ to obtain the corresponding eigenvectors.

$$Te_i = \lambda_i e_i \Rightarrow Te_i - \lambda_i e_i = 0 \Rightarrow (T - \lambda_i I)e_i = 0$$

$$(T - \lambda I)\vec{v} = \vec{0}$$

$$\boxed{\det(T - \lambda I) = 0}$$

Effect of H on the unit circle:

$$H e_1 = 3 e_1 \quad \text{and} \quad H e_2 = -1 e_2$$

$$\underline{e_1} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \underline{e_2} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$H(\underbrace{\alpha_1 e_1 + \alpha_2 e_2}_v) = \alpha_1 \lambda_1 e_1 + \alpha_2 \lambda_2 e_2$$

$$\begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}$$

